क्षिति Kshiti

BHUMI VIKAS SAHAYAK · भूमि विकास सहायक

Building Permission Counter · Bhopal Municipal Corporation · Government of Madhya Pradesh

PROJECT SUMMARY · परियोजना सारांश

THETA172 · TRACK 1

11-10-2026 15:37 IST

IN ONE LINE · एक पंक्ति में

Kshiti answers building-permission questions on the MP Bhumi Vikas Rules 2012 at the Bhopal Municipal Corporation counter, in English or Hindi, with the rule number and the exact text it came from. When the rules are silent, it says so.

THE USER AND THE PROBLEM · उपयोगकर्ता और समस्या

User: a building-permission clerk at the BMC counter. A citizen asks: "My plot is 200 sq m on a 9 m road. How much gap do I leave at the front, how many floors can I build, and what fee do I pay?" The answers sit in tables across a 108-page rulebook that keeps changing, and questions come in English and Hindi. The clerk needs a correct answer within a minute, with the rule number to show the citizen.

WHAT KSHITI DOES · क्षिति क्या करता है

1. Reads the question in milliseconds: language, topic, the citizen's numbers.
2. Calculates figures in code from tables checked against the official PDF: front, rear and side open space (Rule 56), FAR and maximum built-up area (Rule 61), height band (Rule 42), fees (Rule 21).
3. Writes a short answer in the question's language from the rules only, quoting the exact paragraph: Claude Opus 5.5 when an Anthropic key is set, gpt-oss-120b on Groq otherwise (the engine used at the event).
4. Checks every quote word for word against the official text; removes anything that does not match; flags rule numbers that do not exist; flags the 28-08-2026 amendment.
5. Shows the proof (rule, page, official text, PDF link) and prints a PDF slip with a receipt ID and QR code.

EXAMPLE: THE COUNTER QUESTION · उदाहरण

Item · मद	Value · मान	Rule · Page	Table row (official)
Front open space सामने का खुला स्थान	**3.0 m**	Rule 56(1)(a) Page 77	`Up to 9.0 meters`
Max built-up area अधिकतम निर्मित क्षेत्र	**250 sq m (FAR 1.25)**	Rule 61 (Table 8) Page 83	`Resdential  1.25`
Maximum height अधिकतम ऊँचाई	**12.5 m**	Rule 42(1) (Table 4) Page 58	`12.0 m. 1000 sq. 18 m.  1:1.50`
Fees (at 250 sq m) शुल्क	**Rs 3,250**	Rule 21(2)(ii) + 21(3)(b) Page 36	`201  300  3,000`

Rules used: MP Bhumi Vikas Rules 2012, consolidated to 06-03-2025 (MP Town and Country Planning). Rulebook fingerprint SHA-256 e9bfb8e24b9663546b3ae3e4...

For information only. Kshiti points to the rule; the competent authority decides. Not legal advice. केवल जानकारी हेतु; अंतिम निर्णय सक्षम प्राधिकारी का।

WHY IT CAN BE TRUSTED · भरोसा क्यों

- Numbers from code, not from the language model; every table row tested against the rule text.
- Quotes verified word for word; answers with no verified quote are marked "Not verified".
- "Not found" when the rules are silent; never answers from general knowledge.
- Never assumes a missing fact (road width, plot area, height); asks the clerk.
- Every answer and slip states the rules version (consolidated 06-03-2025) and a SHA-256 fingerprint.
- Numbers read from the question are accepted only if written in it; special uses never get residential figures.

MEASURED ACCURACY · मापी गई सटीकता

Live run on the 17 official sample questions (openai/gpt-oss-120b on Groq, 11-10-2026 15:35): correct rule 15/15, "not found" 2/2 (wrong not-found 0), table figures 5/5, answer language 17/17, quotes verified 37/37, invented rule numbers 0, amendment flagged 1/1. Q14 was answered by the calculator after the model wrongly said "not found". Single answers took 3 to 5 s; the batch median was higher because of free-tier rate limits.

KNOWN LIMITS · ज्ञात सीमाएँ

Rules consolidated to 06-03-2025; only the known 28-08-2026 amendment is flagged, and its Gazette text is not available. Calculator covers residential setbacks, FAR, height band and fees. The rules give no number of floors. Source quotes are from the English PDF. A town's Development Plan may differ (Rule 103). Not legal advice.

RUN IT · TEAM · चलाएँ · टीम

https://github.com/Priyanshu12116/KSHITI
pip install -r requirements.txt · add keys to .env · python run.py · http://localhost:8000
Team THETA172 · Priyanshu Sharma (solo) · Track 1, Claude Impact Lab × Cerebras, Bhopal, 11-10-2026 · MIT License